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Alkyl-Capped Block Copolymer Surfactants for Remediation of Soil Water Repellency and Heterogeneous Rootzone Moisture

ABSTRACT: Soil water repellency (SWR) leads to nonwetttable soils, heterogeneous rootzone moisture, reduced irrigation efficiency, leaching due to preferential flow, localized dry spots, and deterioration of turfgrass quality. Nonionic hydroxyl-terminated methyl oxirane-oxirane (EO/PO triblock copolymers) copolymers are commonly employed to remediate SWR. To determine if molecular modification influenced remediation performance, C₁–C₄ alkyl ethers of methyl oxirane-oxirane copolymers were produced through etherification of hydroxyl-terminated methyl oxirane-oxirane copolymers. ACA 1820, an alkyl ether of methyl oxirane-oxirane copolymer (MW approximately 2500) (applied monthly at 1.9 kg ha⁻¹) reduced SWR and maintained quality at higher levels ($p=0.05$) for longer periods than the hydroxyl-terminated methyl oxirane-oxirane copolymer or the control. Turf density in plots treated with the alkyl ether of methyl oxirane-oxirane copolymer was three-fold greater than the control and two-fold greater than in plots treated with the hydroxyl-terminated methyl oxirane-oxirane copolymer. Rootzone moisture was more homogeneous in plots treated with ACA1820 than other treatments. Differences in visual turf quality were observed for up to nine months after the last application of the alkyl ether of methyl oxirane-oxirane copolymer. The observed effects are proposed to be due to interfacial interactions of the alkyl caps both in solution and at hydrophobic soil surfaces.

Introduction

Soil water repellency impacts a range of soil physical processes, including infiltration, preferential flow, and the distribution and dynamics of soil moisture within the rootzone [1–3]. These perturbations influence irrigation efficiency, plant productivity, performance of fertilizers and pesticides, leaching, and runoff [4,5].

When water is applied through precipitation or irrigation, surface runoff and preferential flow are exacerbated leading to leaching of aqueous compositions containing pesticides and fertilizers into ground and surface waters. Water repellency may significantly impact the effective infiltration of water into affected soils, rootzone water reserves, and turf productivity [6–8].

Water repellency of a soil is not only a function of the initial soil composition water content, but is also a function of soil particle size (sands are more prone to water repellency than clays), as well as the type and amount of soil organic matter associated with it [3,9–12]. Soil organic matter may be free and occur in interstitial spaces or adsorbed to and coating soil particle surfaces. Soil organic substances may originate from fungal hyphae, humic substances, or decomposing plant material (leaf waxes, clippings, dead roots, thatch). As soils pass through repeated wet to dry cycles, adsorbed organic matter undergo conformational changes to form nonpolar, hydrophobic films on soil particle surfaces or may be present on interstitial organic matter [13].

Soil water repellency is a well established phenomenon in soils supporting highly managed turfgrass stands and has been reviewed extensively [9,12–14]. On newly constructed golf courses,

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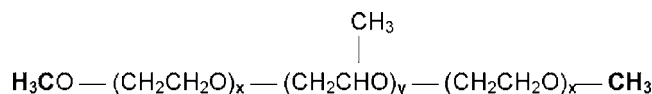


FIG. 1—Structure of a methyl-terminated, alkyl ether of methyl oxirane-oxirane copolymer.

this phenomenon develops rapidly (usually within three years) with visible symptoms occurring seasonally under periods of high evaporative demand. Symptoms include turf wilting and development of dry areas, often impervious to water. These water repellent areas (referred to as localized dry spots or dry patch) are associated with degrading organic matter of plant or microbial origin.

Nonionic surfactants are commonly used for amelioration of dry spot symptoms and to improve the wetting and infiltration in water repellent soils [6–8,15–19]. Over the past decade, nonionic hydroxyl-terminated methyl oxirane-oxirane copolymers [ethylene oxide-propylene oxide (EO/PO) tri-block copolymer surfactants] have become established as the backbone of surfactants used for the amelioration of SWR in golf course soils supporting intensively managed turfgrass.

While these compounds remediate SWR and reduce localized dry areas, other factors including homogeneity of effects, unsatisfactory drydown, and differences in performance based on formulation have been documented [16,19,20]. A search was initiated to address these deficiencies in surfactant performance, with the objective of identifying and developing novel compounds that were more effective in reducing SWR, produced more homogeneous wetting and drydown in the upper portions of the rootzone, and increased turf productivity. The specific studies presented in this report are illustrative of a greater body of research conducted worldwide over four years. The surfactant technology in this report is the subject of pending patent applications in the United States and worldwide [21].

Materials and Methods

Preparation of Alkyl Ethers of Methyl Oxirane-oxirane Copolymers

C₁–C₄ alkyl ethers of methyl oxirane-oxirane copolymers were prepared by BASF-AG, Ludwigshafen, Germany, by reacting a hydroxyl-terminated methyl oxirane-oxirane copolymer (MW approximately 2500) with sodium hydroxide and methyl chloride or dimethyl sulfate [22]. Etherification resulted in a by-product salt removed via filtration, decantation, extraction, or distillation. The molecule used in this study contained two end-capped methyl groups, substituting the terminal hydroxyl groups of the starting methyl oxirane-oxirane co-polymer (Fig. 1).

Preparation of Synthetic Water Repellent Sand

Synthetic water repellent sand was prepared by coating hydrophilic greens mix sand (Unipar Top-dressing Sand [TD-310] UniMin Corporation, New Canaan, CT) with octadecyltrichlorosilane (OTS) [2]. Water repellency in OTS-treated sand was determined using the water drop penetration time (WDPT) test [23]. In this test, sand was evenly loaded into a petri dish and the surface smoothed to produce a flat, unconsolidated substrate surface. Drops of distilled water (30 μL) were placed on the smoothed surface of the sample and the time that elapsed for the drops to completely infiltrate was determined. Tests were conducted at approximately 20°C.

Column Studies

Surfactants evaluations were conducted in the laboratory using a standardized column test [24] and OTS-treated water repellent sand. Columns (5 mm ID) were filled to a height of 60 cm with water repellent sand and the time for the advancing wetting front of an equivalent volume of water or aqueous surfactant solutions (4000, 6000, and 8000 mg L⁻¹) to infiltrate through the water repellent soil column was measured. The initial hydraulic head for each test solution was 60 mm H₂O. Tests

TABLE 1—Treatments used for evaluation of hydroxyl-terminated methyl oxirane-oxirane copolymer and alkyl ether of methyl oxirane-oxirane copolymer surfactants for soil water repellency reduction.

Treatment	Application Rate [kg ha ⁻¹]	Water [L ha ⁻¹]
Control	0	800
Hydroxyl-terminated	1.94	800
Alkyl-terminated	0.65	800
Alkyl-terminated	1.29	800
Alkyl-terminated	1.94	800
Alkyl-terminated	2.59	800

were conducted at approximately 20°C and a relative air humidity of approximately 50 %. Four replicate columns were evaluated with each treatment by dilution combination.

Field Evaluations

The sites selected for evaluation consisted of water repellent soils. Each location was a highly maintained turf area where standard cultural practices were maintained.

Turf Quality and Water Repellency Management

Evaluations were conducted on a sand-based research green built to USGA specifications and with a history of severe SWR. The test design was a randomized complete block with four replicates per treatment. Turfgrass (*Agrostis stolonifera*) was maintained at greens height. Treatments (Table 1) were applied using a pressure-powered sprayer or a volume-controlled hydraulic sprayer. Control plots received water only. Treatments were applied monthly for four months beginning prior to the onset of seasonally high evaporative demand.

Visual quality evaluations were conducted based on a scale of 1–9 (1=dead, brown turf, 9=healthy, dark green turf). Assessments were made prior to treatment and approximately every 7–14 days thereafter.

To determine SWR, soil cores were collected prior to treatment and then every two to four weeks after subsequent applications for 20 weeks. Five soil cores were collected every two weeks from each plot. Soil cores were sectioned into 1 cm increments to a depth of 4 cm. Intact soil cores were dried at ambient laboratory conditions (approximately 20°C and a relative air humidity of approximately 50 %). SWR (persistence of actual water repellency) was measured on intact soil cores using the WDPT test.

Soil Water Content

Soil volumetric water content was monitored over two growing seasons in a severely water repellent sandy soil on a golf course fairway. Three treatments were compared in 1.5 m by 3.5 m plots: hydroxyl-terminated methyl oxirane-oxirane copolymer (1.94 kg ha⁻¹), alkyl ether of methyl oxirane-oxirane copolymer (1.94 kg ha⁻¹), and an untreated control. Treatments were applied every three to four weeks with a pressurized sprayer equipped with a flat-fan nozzle.

Volumetric soil water contents at the 0–5-cm depth were measured using a hand-held time domain reflectrometry device (Trime TDR Probe, IMKO, Ettlingen, Germany) in a 10 cm by 10 cm grid over an area 0.9 m by 0.9 m. Measurements were made periodically over two growing seasons.

Turf Density

Turf density was compared in September 2003. A section of each plot 50 cm by 50 cm in size was divided into a total of 25 dm² subplots and individual plants in each dm² subplot counted.

TABLE 2—Comparison of infiltration times (in minutes) for different surfactant solutions to infiltrate a 6-cm column of synthetically produced water repellent sand.

Surfactant	Infiltration Time (min)		
	Surfactant Concentration (mg L ⁻¹)		
	4000	6000	8000
Hydroxyl-terminated	97.3 a ^a	62.3 a	44.8 a
Alkyl-terminated	76.3 b	55.3 b	24.6 b

^aNumbers in columns followed by the same letter are not significantly different according to Duncan's multiple range test ($p=0.05$).

Statistical analyses

All treatments were analyzed using an Analysis of Variance model procedure and treatment means were separated using Duncan's Multiple Range Test or Least Significant Difference (LSD) Test.

Results

Column Studies

The WDPT values obtained with the OTS-treated sand exceeded 6 h and may be categorized accordingly as a severely water repellent soil [8]. When evaluated in a column study, the time for water to infiltrate the 6-cm column of OTS-treated, water repellent sand was in excess of 176 h, confirming the categorization of these soils as severely water repellent. Substitution of the two terminal hydroxyl groups with methyl groups had a profound effect on infiltration time (Table 2). At each of the surfactant concentrations evaluated, infiltration times for the alkyl ether of the methyl oxirane-oxirane copolymer were significantly shorter than for the hydroxyl-terminated oxirane-oxirane copolymer ($p=0.05$). The substitution of methyl groups for the terminal hydroxyl groups improved the speed of infiltration potentially by modifying the aggregate level of hydrophobicity of the alkyl ether of the methyl oxirane-oxirane copolymer.

Turf Quality

Prior to initial treatment applications no differences were observed between plots (Table 3). Statistically significant differences in visual quality were observed between selected surfactant treatments and the control within one month of treatment applications. Statistically significant differences in turf quality appeared between the control plots and the plots treated with the hydroxyl-terminated methyl oxirane-oxirane copolymer after two months. Turf quality ratings in plots treated with the alkyl ether of methyl oxirane-oxirane copolymer applied at the two lowest rates (0.65 kg or 1.29 kg ha⁻¹) were statistically equivalent to the hydroxyl-terminated methyl oxirane-oxirane copolymer applied at 1.94 kg ha⁻¹. Quality ratings in plots treated with the alkyl ether of methyl oxirane-oxirane copoly-

TABLE 3—Effect of soil surfactant treatment on bentgrass visual turfgrass quality rating.

Date	6/8	6/9	6/11	6/15	6/22	7/6	7/20	8/3	8/17	8/31	9/14	9/28	10/12	10/26
Control	7.9 ^a	7.8	7.5	7.4	6.9	6.3 b	6.5 b	6.1 c	5.5 d	5.6 d	5.4 d	5.4 d	5.6 e	5.6 c
Hydroxyl-terminated @ 1.94 kg ha ⁻¹	7.5	7.4	7.4	7.5	7.2	6.8 ab	6.9 ab	6.9 ab	6.8 bc	6.5 bc	6.8 ab	6.6 c	6.4 cd	6.4 b
Alkyl-terminated @ 0.65 ka ha ⁻¹ 1	7.5	7.5	7.5	7.5	7.3	6.9 ab	6.9 ab	6.8 b	6.4 c	6.3 c	6.6 bc	6.4 c	6.3 d	6.3 b
Alkyl-terminated @ 1.29 ka ha ⁻¹	7.5	7.6	7.6	7.6	7.1	7.3 a	6.9 a	6.9 ab	6.5 c	6.8 abc	6.8 ab	6.8 bc	6.6 bcd	6.4 b
Alkyl-terminated @ 1.94 kg ha ⁻¹	8.0	8.0	7.8	7.5	7.1	7.0 a	7.1 a	7.4 a	7.4 a	7.1 a	7.1 a	7.4 a	7.0 ab	6.9 ab
Alkyl-terminated @ 2.59 kg ha ⁻¹	7.6	7.6	7.5	7.8	7.3	7.0 a	7.1 a	7.1 ab	7.3 a	7.0 ab	7.1 a	7.3 ab	7.3 a	7.1 a

^aNumbers in columns followed by the same letter are not significantly different according to Duncan's multiple range test ($p=0.05$).

TABLE 4—Effect of soil surfactant treatment on mean soil water repellency (WDPT) in a water repellent soil receiving four monthly surfactant treatments.

Treatment	Rate [kg ha ⁻¹]	Depth (cm)	
		0.1	1.0
Control	0	268 d ^a	215 d
Hydroxyl-terminated	1.94	109 bc	178 c
Alkyl-terminated	0.63	137 c	158 bc
Alkyl-terminated	1.29	80 ab	155 bc
Alkyl-terminated	1.94	56 a	77 a
Alkyl-terminated	2.59	58 a	68 a
LSD (<i>p</i> =0.1)		52.6	67.5

^aNumbers in columns followed by the same letter are not significantly different according to Least Significant Difference test (*p*=0.1).

mer applied at 1.94 kg ha⁻¹ to 2.59 kg ha⁻¹ could be separated from the control after one month. Over the remainder of the test period (a period of nearly four months), these two treatments tended to be significantly higher (*p*=0.05) than the control and the hydroxyl-terminated methyl oxirane-oxirane copolymer.

Soil Water Repellency

Water repellency predominated in the top of the soil profile, the region containing the highest organic matter content. No statistically significant differences in SWR were found between plots prior to treatment applications (data not presented). Soil water repellency in the untreated control plots was severe over the four-month evaluation period (Table 4). The hydroxyl-terminated methyl oxirane-oxirane copolymer significantly reduced SWR at both test depths as compared to the untreated control (*p*=0.1). Applications of the alkyl ether of methyl oxirane-oxirane copolymer at rates of as low as 0.65 kg ha⁻¹ reduced SWR in treated soils. Soil water repellency reductions by the low rate alkyl ether of methyl oxirane-oxirane copolymer treatment provided statistically equivalent SWR control to the hydroxyl-terminated methyl oxirane-oxirane copolymer applied at a three-fold higher treatment rate (1.94 kg ha⁻¹). As application rate of the alkyl ether of methyl oxirane-oxirane copolymer increased, SWR decreased at both test depths in the rootzone (Table 4). When applied at the same rates (1.94 kg ha⁻¹), the alkyl ether of methyl oxirane-oxirane copolymer provided SWR reductions superior to that of the hydroxyl-terminated methyl oxirane-oxirane copolymer.

Soil Water Content

Rootzone soil moisture homogeneity was greatest in soils treated with the alkyl ether of methyl oxirane-oxirane copolymer (Fig. 2). Conversely, soil moisture heterogeneity increased in plots treated with the hydroxyl-terminated methyl oxirane-oxirane copolymer (1.94 kg ha⁻¹) and the control, respectively. Other differences were also observed between the treatments. Untreated soils had the lowest soil volumetric water content while water contents in surfactant treated plots were consistently higher. Soil volumetric water contents were generally lower and in tighter ranges in plots treated with the alkyl ether of methyl oxirane-oxirane copolymer than in the plots treated with the hydroxyl-terminated methyl oxirane-oxirane copolymer (data not presented). These results suggest that the alkyl ether of methyl oxirane-oxirane copolymer produced a thinner, more continuous film of water in treated soils.

TABLE 5—Changes in plant density changes in turfgrass growing in water repellent soils treated with surfactants.

Treatment [1.94 kg ha ⁻¹]	Plants dm ²
Untreated control	107a ^d
Hydroxyl-terminated	229 b
Alkyl-terminated	340 c

^dNumbers in columns followed by the same letter are not significantly different according to Duncan's multiple range test (*p*=0.05).

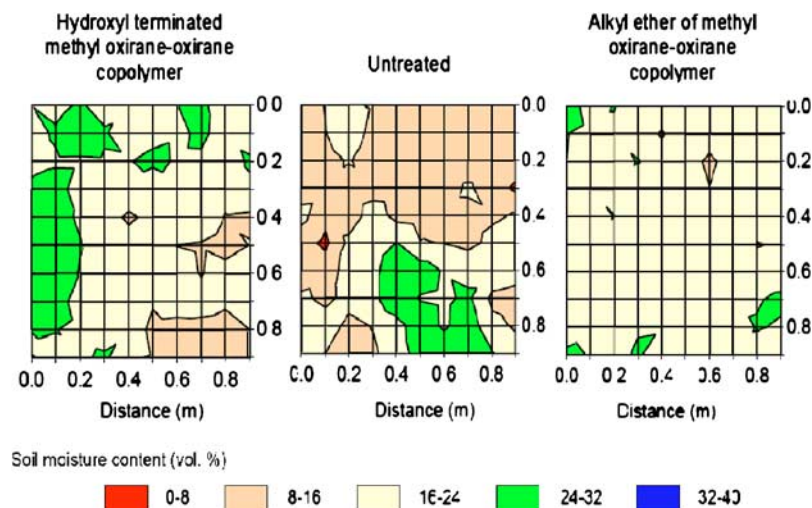


FIG. 2—Soil volumetric water content heterogeneity in water repellent soils treated with hydroxyl-terminated or alkyl ethers of methyl oxirane-oxirane copolymer.

Turf Density

As soil volumetric water content became more homogeneous and summer stress conditions were imposed, turf performance increased. Under these conditions, the alkyl ether of methyl oxirane-oxirane copolymer at the 1.94 kg ha^{-1} treatment rate maintained turfgrass quality at statistically higher levels ($p=0.05$) for longer periods than the hydroxyl-terminated methyl oxirane-oxirane copolymer or the control (Table 5). The alkyl ether of methyl oxirane-oxirane copolymer increased turfgrass density in treated soil three-fold over the control while the hydroxyl-terminated methyl oxirane-oxirane copolymer increased density two-fold over the control. Increased plant density equates with more shoots per unit area and better transpiration and evaporative cooling of the turf canopy.

Discussion

Substitution of the two terminal hydroxyl groups with methyl groups had a profound effect on performance time of the alkyl ether of methyl oxirane-oxirane copolymer surfactant. The alkyl ether of methyl oxirane-oxirane copolymer accelerated infiltration through a column of water repellent soil, reduced SWR more effectively, and increased turfgrass quality and density under periods of high evaporative demand. Even during periods of high evaporative demand, soils treated with the alkyl ether of methyl oxirane-oxirane copolymer distributed water more homogeneously in the rootzone than could be achieved with the hydroxyl-terminated methyl oxirane-oxirane copolymer.

The observed effects are hypothesized to be due to unique interfacial interactions of the C_1 – C_4 alkyl caps both in solution and at hydrophobic surfaces. The alkyl caps increase the lipophilic (hydrophobic) nature of the surfactant causing the molecule to be less soluble in the water phase. When in solution, these molecules will have a greater propensity to accumulate at the liquid-air interface. When coming in contact with a hydrophobic (lipophilic) surface, the molecule will migrate preferentially to this surface and both the methyl oxirane moiety and the alkyl caps will adsorb. The addition of the alkyl caps should change the orientation of the adsorbed surfactant molecule on the hydrophobic surface, thus increasing the surface area influenced by the hydrophilic moieties of the molecule. Increased surfactant adsorption and modification of the orientation of the hydrophilic moieties to a more intimate association with the hydrophobic solid surface leads to a more uniform film of water and increased accessibility to capillary water flow.

As a consequence of improved rootzone hydrological status, turfgrass roots may more effectively access water during periods of elevated evaporative demand and growth processes can be maintained under stress conditions. Even under severe water limitations, soils treated with alkyl ether of methyl

oxirane-oxirane copolymer did not fall below the critical water content—the point at which soils become nonwetttable. Treated soils readily accepted applied water, even when dry. Better control of soil water within tighter ranges produced predictable, homogeneous infiltration and homogeneous moisture distribution, resulting in homogeneous drainage, drydown, and aeration. The test results suggest that the alkyl ether of methyl oxirane-oxirane copolymer by improving rootzone hydrological status may provide better access to pores and thus better access to macro- and micro-nutrients and potentially to the viability of soil microorganisms active in mineralization. As rootzone hydrological conditions became more homogeneous, an environment was created for better turf performance and for better control of playing conditions resulted.

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References

- [1] Dekker, L. W. and Ritsema, C. J., “How Water Moves in a Water Repellent Sandy Soil: Potential and Actual Water Repellency,” *Water Resour. Res.*, Vol. 30, 1994, pp. 2507–2517.
- [2] Bauters, T. W. J., DiCarlo, D. A., Steenhuis, T. S., and Parlange, J-L., “Preferential Flow in Water Repellent Sands,” *Soil Sci. Soc. Am. J.*, Vol. 62, 1998 pp. 1185–1190.
- [3] DeBano, L. F., “Historical Overview of Soil Water Repellency,” *Oil Water Repellency—Occurrence, Consequences, and Amelioration*, C. J. Ritsema and L. W. Dekker, Eds., Elsevier, Amsterdam, The Netherlands, 2003, pp. 3–21.
- [4] Doerr, S. H., Shakesby, R. A., and Walsh, R. P. D., “Soil Water Repellency: Its Causes, Characteristics and Hydro-geomorphological Significance,” *Earth-Sci. Rev.*, Vol. 51, 2000, pp. 33–65.
- [5] Clothier, B. E., Vogeler, I., and Magesan, G. M., “Solute Transport Through a Hydrophobic Soil,” *Soil Water Repellency—Occurrence, Consequences, and Amelioration*, C. J. Ritsema and L. W. Dekker, Eds., Elsevier, Amsterdam, The Netherlands, 2003, pp. 225–234.
- [6] Dekker, L. W., Doerr, S. H., Oostindie, K., Ziogas, A. K., and Ritsema, C. J., “Water Repellency and Critical Soil Water Content in a Dune Sand,” *Soil Sci. Soc. Am. J.*, Vol. 65, 2001, pp. 1667–1674.
- [7] Leinauer, B., Rieke, P. E., VanLeeuwen, D., Sallenave, R., Makk, J., and Johnson, E., “Effects of Soil Surfactants on Water Retention in Turfgrass Rootzones,” *Int. Turfgrass Soc. Res. J.*, Vol. 9, 2001 pp. 542–547.
- [8] Dekker, L. W., Oostindie, K., Kostka, S. J., and Ritsema, C. J., “Effects of Surfactant Treatment on the Wettability of a Water Repellent Grass-covered Dune Sand,” *Austral. J. Soil Res.*, Vol. 43, 2005, pp. 383–395.
- [9] Wilkinson, J. F. and Miller, R. H., “Investigation and Treatment of Localized Dry Spots on Sand Golf Greens,” *Agron. J.*, Vol. 70, 1978, pp. 299–304.
- [10] de Jonge, L. W., Jacobsen, O. H., and Moldrup, P., “Soil Water Repellency: Effects of Water Content, Temperature, and Particle-size,” *Soil Sci. Soc. Am. J.*, Vol. 63, 1999, pp. 437–442.
- [11] Hallett, P. D., Ritz, K., and Wheatley, R. E., “Microbial Derived Water Repellency in Golf Course Soil,” *Int. Turfgrass Soc. Res. J.*, Vol. 9, 2001, pp. 518–524.
- [12] Karnok, K. J. and Tucker, K. A., “Water Repellent Soils Part I. Where Are We Now?,” *Golf Course Management*, Vol. 70, No. 6, 2002, pp. 59–62.
- [13] Doerr, S. H., Llewellyn, C. T., Douglas, P., Morley, C. P., Mainwaring, K. A., Haskins, C., Johnsey, L., Ritsema, C. J., Stagnitti, F., Allison, G., Ferriera, A. J. D., Keizer, J. J., Ziogas, A. K., and Diamantis, J., “Extraction of Compounds Associated with Water Repellency in Sandy Soils of Different Origin,” *Austral. J. Soil Res.*, Vol. 43, 2005, pp. 225–237.

- [14] Karnok, K. J. and Tucker, K. A., "Water Repellent Soils. Part II. More Questions and Answers," *Golf Course Management*, Vol. 70, No. 7, 2002, pp. 49–52.
- [15] Wallis, M. G. and Horne, D. J., "Soil Water Repellency," *Adv. Soil Sci.*, Vol. 20, 1992, pp. 91–146.
- [16] Cisar, J. L., Williams, K. E., Vivas, H. E., and Haydu, J. J., "The Occurrence and Alleviation by Surfactants of Soil Water Repellency on Sand-based Turfgrass Systems," *J. Hydrol.*, Vol. 231–232, 2000, pp. 352–358.
- [17] Kostka, S. J., "Amelioration of Water Repellency in Highly Managed Soils and the Enhancement of Turfgrass Performance through the Systematic Application of Surfactants," *J. Hydrol.*, Vol. 231–232, 2000, pp. 359–368.
- [18] Karnok, K. J. and Tucker, K. A., "Wetting Agent Treated Hydrophobic Soil and Its Effect on Color, Quality and Root Growth of Creeping Bentgrass," *Int. Turfgrass Soc. Res. J.*, Vol. 9, 2001, pp. 537–541.
- [19] Miller, C., "Comparing Wetting Agents: Long-term Versus Short-term," *Golf Course Management*, Vol. 69, No. 4, 2001, pp. 60–64.
- [20] Miller, C. and Kostka, S. J., "The Effect of Primer 604 Nonionic Surfactant on Water Repellency in Sand-based Soils," *Proceedings of the 5th International Symposium on Adjuvants for Agrochemicals*, Vol. 1, 1998, pp. 291–297.
- [21] Kostka, S. J. and Schuermann, G., "Enhancing Plant Productivity by Improving the Plant Growth Medium Environment with Alkyl Ethers of Methyl Oxirane-oxirane Copolymer Surfactants," Patent Appl. US2005/0221992, WO2005/096816, 2005.
- [22] Birnbach, S., Mueller, J., Wulz, K., and Houben, J., "Etherification of Polyoxyalkylene Derivatives," U.S. Patent No. 4,922,029, 1990.
- [23] Lety, J., "Measurement of Contact Angle, Water Drop Penetration Time and Critical Surface Tension," *Proceedings, Symposium on Water Repellent Soils*, University of California, Riverside, CA, 1969, pp. 43–47.
- [24] Mane, S., Moore, D., and Moore, R. A., "A Simple Method for Determining the Initial and Residual Effectiveness of Soil Wetting Agents," *Int. Turfgrass Soc. Res. J.*, Vol. 7, 1993, pp. 485–488.